

TVEITSUND, Norway

WMO: 014550

Lat: 59.027N

Lon: 8.521E

Elev: 252

StdP: 98.33

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.3	-13.0	-17.3	0.8	-13.9	-15.0	1.0	-12.1	6.8	4.8	5.7	3.7	0.8	30	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	25.3	16.2	23.4	15.5	21.5	14.8	17.6	22.0	16.7	20.9	15.9	19.9	2.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.7	18.8	15.1	11.1	17.9	14.3	10.5	17.2	50.4	22.4	47.7	20.9	45.3	19.9	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.2	4.4	3.8	-17.9	28.1	4.0	1.8	-20.7	29.4	-23.1	30.4	-25.3	31.4	-28.2	32.7		
(5)			-17.0	18.9	4.0	1.1	-19.9	19.7	-22.2	20.4	-24.5	21.0	-27.4	21.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-2.1	-2.6	0.3	4.8	9.4	13.5	15.8	14.8	11.1	6.1	2.0	-1.1
	DBStd	7.49	4.81	5.17	4.28	3.11	3.49	2.63	2.62	2.38	2.58	3.20	3.81	4.87
	HDD10.0	1977	376	353	302	158	55	3	0	1	18	126	241	344
	HDD18.3	4499	634	587	560	407	278	146	88	111	216	378	491	603
	CDD10.0	532	0	0	0	2	35	108	179	151	52	6	0	0
	CDD18.3	13	0	0	0	0	0	2	9	2	0	0	0	0
	CDH23.3	173	0	0	0	0	11	37	104	19	1	0	0	0
	CDH26.7	19	0	0	0	0	1	3	14	1	0	0	0	0
(14) Wind	WSAvg	1.7	1.7	1.7	1.7	1.7	1.7	1.8	1.6	1.6	1.6	1.6	1.7	1.8
(15) Precipitation	PrecAvg	1074	100	60	56	61	73	86	98	121	109	120	109	80
	PrecMax	1631	241	194	141	159	203	185	235	229	356	301	463	153
	PrecMin	815	16	9	9	6	4	9	14	10	41	20	25	16
	PrecStd	188	56	42	33	39	47	43	52	55	67	68	87	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	9.2	13.1	18.4	24.8	26.8	28.5	25.6	20.8	15.6	12.5	10.6
		MCWB	5.8	5.6	7.0	10.1	14.6	15.6	16.9	17.4	14.5	11.8	9.2	7.6
	2%	DB	6.9	7.3	10.1	14.8	21.1	23.9	26.1	23.7	18.7	13.5	10.8	8.6
		MCWB	4.5	4.5	5.6	8.3	12.6	14.5	16.4	16.8	13.4	10.8	8.8	6.4
	5%	DB	5.5	5.9	8.1	12.2	18.5	21.5	24.1	21.7	17.1	12.4	9.1	7.1
		MCWB	3.6	3.3	4.3	7.1	11.6	13.6	15.9	16.0	12.9	10.3	7.2	5.2
	10%	DB	4.3	4.4	6.4	10.3	16.1	19.6	21.8	20.0	15.6	11.3	7.6	5.6
		MCWB	2.6	2.4	3.3	6.0	10.4	12.8	15.3	14.8	12.3	9.4	5.9	3.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	6.4	7.7	11.1	15.5	17.0	18.8	18.7	16.0	13.3	10.6	8.1
		MCDB	8.1	8.8	12.0	15.5	22.3	22.8	24.6	22.9	18.6	14.3	11.6	9.7
	2%	WB	5.1	4.9	6.1	9.1	13.8	15.8	17.7	17.7	14.7	11.8	9.0	6.9
		MCDB	6.7	6.8	9.3	13.2	19.1	21.2	22.9	21.4	16.7	12.8	10.4	8.4
	5%	WB	3.9	3.7	4.8	7.9	12.4	14.8	16.8	16.8	13.9	10.9	7.6	5.5
		MCDB	5.2	5.4	7.4	11.5	16.9	19.7	21.5	20.4	15.8	11.9	9.0	6.9
	10%	WB	2.8	2.6	3.8	6.8	11.2	13.8	16.0	15.9	13.1	9.7	6.2	4.1
		MCDB	4.2	4.2	6.0	9.6	15.1	18.0	20.3	19.1	14.9	11.0	7.5	5.4
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	6.1	7.8	8.8	9.8	9.9	9.3	8.6	7.3	6.0	4.8	5.1
		MCDBR	6.5	7.2	10.7	13.3	14.7	14.3	13.9	11.4	10.3	7.4	6.6	6.7
		MCWBR	5.3	5.2	6.6	7.5	7.4	6.6	6.1	5.5	6.1	5.6	5.7	5.6
	5% WB	MCDBR	5.9	6.5	9.3	11.8	12.4	11.7	10.9	9.3	7.6	5.8	5.9	6.0
		MCWBR	5.2	5.0	6.1	7.2	6.7	6.1	5.4	5.1	5.3	5.0	5.4	5.4
(40) Clear-Sky Solar Irradiance	taub		0.255	0.291	0.316	0.359	0.356	0.343	0.363	0.359	0.332	0.315	0.275	0.232
	taud		2.227	2.308	2.332	2.332	2.431	2.486	2.443	2.447	2.520	2.494	2.326	2.156
	Ebn at Noon		573	737	826	845	876	893	863	839	810	711	550	463
	Edh at Noon		43	67	88	105	102	98	101	93	74	57	40	31
(44) All-Sky Solar Radiation	RadAvg		0.34	0.95	2.43	3.95	5.04	5.77	5.15	4.02	2.61	1.20	0.44	0.23
	RadStd		0.07	0.20	0.21	0.56	0.60	0.44	0.59	0.28	0.34	0.21	0.12	0.03

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page